

PROJECT DIARY

BCAS – SCHOOL OF COMPUTING DEPARTMENT OF COMPUTING BTEC HND PROGRAMME

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Assessor Name: Eng. Sajith

Batch: 22

Project Progress Daily Sheet

Week Start Date	03.08.2025
Week Ending Date	09.08.2025

DAY	ACTIVITY
Monday	Researched IoT applications in safety systems and hazard detection.
Tuesday	Delivered Proposal presentation
Wednesday	
Thursday	Identified IoT architecture models (three-layer, four-layer, five-layer).
Friday	
Saturday	
Sunday	Reviewed ESP32 microcontroller features (Wi-Fi, GPIO pins, memory).

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed: Delivered Proposal presentation

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

09.08.2025

DATE

PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	10.08.2025
Week Ending Date	16.08.2025

DAY	ACTIVITY
Monday	
Tuesday	Created circuit design for sensor, buzzer, and LED integration.
Wednesday	
Thursday	Procured required hardware components (ESP32, MQ-2, flame sensor, buzzer, LEDs, jumper wires).
Friday	
Saturday	Assembled MQ-2 and IR flame sensor connections on breadboard.
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed: Assembled MQ-2 and IR flame sensor connections on breadboard.

Problems Occurred:

Faced difficulties in selecting suitable components for the project

Suggestions Made by the Supervisor:

Gave clarification about the hardware components and suggested select specific components.

Decisions Taken:

According to the suggestions of the assessor, selected suitable components.

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

16.08.2025

DATE

PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	17.08.2025
Week Ending Date	23.08.2025

DAY	ACTIVITY
Monday	Configured ESP32 development environment in Arduino IDE.
Tuesday	Installed required libraries (Blynk, Wi-Fi, ESP32 support).
Wednesday	
Thursday	Tested ESP32 Wi-Fi connectivity with simple code.
Friday	
Saturday	Programmed ESP32 for MQ-2 gas sensor analog input reading.
Sunday	Programmed ESP32 for IR flame sensor digital detection.

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed: Programmed ESP32 for MQ-2 and IR flame sensors

Problems Occurred:

Faced difficulties in choosing digital and analog pins.

Suggestions Made by the Supervisor:

Gave clarity of digital and analog pins of the developer boards and the use cases of them.

Decisions Taken:

Selected analog pin for MQ-2 sensor and selected digital pin for IR flame sensor

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

23.08.2025

DATE

PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	24.08.2025
Week Ending Date	30.08.2025

DAY	ACTIVITY
Monday	Integrated buzzer as an alarm actuator triggered by hazard detection.
Tuesday	Configured green LED as safe indicator and red LED as hazard indicator.
Wednesday	
Thursday	Linked ESP32 with Blynk platform using authentication token.
Friday	Set up Blynk dashboard with widgets, notification
Saturday	
Sunday	Conducted initial test of sensor data transmission to Blynk app.

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed: connected the project with Blynk application for real-time alerts and notification

Problems Occurred:

Faced problems while configuring the Blynk auth code with Esp32

Suggestions Made by the Supervisor:

Gave a practical tutorial of how to set up Blynk dashboard

Decisions Taken:

Based on the tutorial perfectly configured Blynk notification functionality

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

30.08.2025

DATE

PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	31.08.2025
Week Ending Date	06.09.2025

DAY	ACTIVITY
Monday	
Tuesday	Performed gas detection tests using controlled smoke/gas sources.
Wednesday	Conducted flame detection tests with small flame sources at safe distance.
Thursday	
Friday	Tested buzzer and LED indicators for correct responses
Saturday	Verified push notifications in Blynk app under hazard conditions.
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed: Tested the Gas detection, flame detection, buzzer and alert notifications

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

06.09.2025

DATE

PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	07.09.2025
Week Ending Date	13.09.2025

DAY	ACTIVITY
Monday	Adjusted MQ-2 threshold to reduce false alarms.
Tuesday	Introduced delay control for buzzer and LEDs.
Wednesday	
Thursday	Gathered user feedback on usability of Blynk app and local indicators.
Friday	
Saturday	Improved system by adding dual LED indicators for real-time status.
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed: Made adjustments on the sensors to ensure the reliability of the system

Problems Occurred:

Constant false alarm occurred frequently

Suggestions Made by the Supervisor:

Insisted on adjusting the threshold of the MQ-2 sensor

Decisions Taken:

Adjusted the MQ-2 threshold and it worked.

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

13.09.2025

DATE

PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	14.09.2025
Week Ending Date	19.09.2025

DAY	ACTIVITY
Monday	Re-tested complete system for stability and accuracy.
Tuesday	
Wednesday	Delivered the Final presentation
Thursday	Completed logbook entries and prepared submission materials.
Friday	
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed: Delivered Final Presentation & report submission

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

19.09.2025

DATE

PROJECT SUPERVISOR'S SIGNATURE